

AI Based Smart System Development (Real Time Hardware Implementation)

Week – 2 Project Assignment-1

- 1) Design a smart distance measurement system using an ESP32 and an HC-SR04 ultrasonic sensor. The measured distance should be continuously displayed on a 16×2 I2C LCD as well as on the Serial Monitor.

The system should also classify the measured distance into three zones:

Distance	Status
Less than 10 cm	DANGER
10–30 cm	CAUTION
Greater than 30 cm	SAFE

Due Date of Submission: 17/07/2026